

MRC Lecture 1 — Derivation Guide 4

The Softmax Function: Converting Scores to Probabilities

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1. Motivation

We now have compatibility scores between agents and targets: dot products (Derivation Guide 1), organized into a similarity matrix (Derivation Guide 2), optionally normalized by L2 normalization (Derivation Guide 3). The remaining question is: **how do we convert a vector of scores into a probability distribution?**

This is the fundamental problem of competitive selection. An antibody in the bloodstream encounters thousands of potential antigens. A transcription factor scans millions of candidate binding sites. A transformer query must distribute attention across all keys in the sequence. In each case, the system has a vector of compatibility scores and must produce a probability distribution over candidates — a vector of non-negative numbers that sum to one, where higher scores correspond to higher probabilities.

The function that performs this conversion is the **softmax function**. In Lecture 6, we will prove that softmax is the *unique* function satisfying five axioms of competitive selection (Luce, 1959). In this guide, we define softmax, compute it by hand, verify its properties, and build intuition for how it behaves.

2. Definition

Definition. Given a score vector $z = (z_1, z_2, \dots, z_M) \in \mathbb{R}^M$, the softmax function produces a probability vector $P = (P_1, P_2, \dots, P_M)$ where:

$$P_j = \text{softmax}(z)_j = \frac{\exp(z_j)}{\sum_{k=1}^M \exp(z_k)}$$

The function has two stages: (1) exponentiate each score, and (2) normalize by the sum of all exponentials.

Notation. The denominator $Z = \sum_{k=1}^M \exp(z_k)$ is called the **partition function** in statistical mechanics and the **normalization constant** in machine learning. In the Boltzmann distribution, Boltzmann called this quantity the Zustandssumme (German: “sum over states”), abbreviated Z — and the letter has been used ever since.

3. Worked Example 1: Basic Computation

Problem. Compute softmax for the score vector $z = (2.0, 1.0, 0.1)$.

Step 1. Exponentiate each score:

$$\begin{aligned}\exp(2.0) &= 7.389 \\ \exp(1.0) &= 2.718 \\ \exp(0.1) &= 1.105\end{aligned}$$

Step 2. Compute the partition function (sum of exponentials):

$$Z = 7.389 + 2.718 + 1.105 = 11.212$$

Step 3. Divide each exponential by Z :

$$P_1 = \frac{7.389}{11.212} = 0.659$$

$$P_2 = \frac{2.718}{11.212} = 0.242$$

$$P_3 = \frac{1.105}{11.212} = 0.099$$

Step 4. Verify:

$$P_1 + P_2 + P_3 = 0.659 + 0.242 + 0.099 = 1.000 \checkmark$$

Interpretation. The input scores were $2.0 > 1.0 > 0.1$. The output probabilities are $0.659 > 0.242 > 0.099$. The rank order is preserved. The highest score (2.0) receives 66% of the probability — more than its proportional share — because the exponential amplifies differences.

4. Properties of Softmax

We verify four properties that will become the first four axioms of competitive selection in Lecture 6.

Property 1: Positivity. For all j : $P_j > 0$.

Proof. The exponential function is strictly positive: $\exp(z_j) > 0$ for all $z_j \in \mathbb{R}$. The denominator $Z > 0$ since it is a sum of positive terms. Therefore $P_j = \exp(z_j) / Z > 0$. No candidate ever receives exactly zero probability. ■

In molecular binding, this corresponds to the physical fact that at non-zero temperature, every molecular interaction has non-zero probability — even energetically unfavorable ones. An antibody can bind a non-cognate antigen; it is just very unlikely.

Property 2: Normalization. $\sum_{j=1}^M P_j = 1$.

Proof.

$$\sum_{j=1}^M P_j = \sum_{j=1}^M \frac{\exp(z_j)}{Z} = \frac{1}{Z} \sum_{j=1}^M \exp(z_j) = \frac{Z}{Z} = 1 \blacksquare$$

The output is a valid probability distribution.

Property 3: Rank preservation (monotonicity). If $z_A > z_B$ then $P_A > P_B$.

Proof. Since $\exp$ is strictly increasing, $z_A > z_B$ implies $\exp(z_A) > \exp(z_B)$. Both share the same denominator Z , so:

$$P_A = \frac{\exp(z_A)}{Z} > \frac{\exp(z_B)}{Z} = P_B \blacksquare$$

Higher scores always produce higher probabilities. The softmax function never reverses the ranking.

Property 4: Unrestricted domain. The function accepts any real-valued input: $z_j \in (-\infty, +\infty)$.

Proof. The exponential $\exp(z_j)$ is defined for all $z_j \in \mathbb{R}$. There is no restriction to non-negative inputs. In molecular binding, energies span the full real line: favorable interactions have negative ΔG (positive scores after the sign flip), and unfavorable interactions have positive ΔG (negative scores). ■

5. The Fifth Property: Independence of Irrelevant Alternatives

There is a fifth property — the most important one — that softmax satisfies and that most other candidate functions do not. This property is called the **independence of irrelevant alternatives (IIA)**, and it is the subject of Lecture 6. We state it here and verify it for softmax, without yet proving uniqueness.

Statement (IIA). The ratio of selection probabilities between any two candidates depends only on their own scores, not on what other candidates are present:

$$\frac{P_A}{P_B} = \frac{\exp(z_A)}{\exp(z_B)} = \exp(z_A - z_B)$$

The third candidate, the fourth candidate, the hundredth candidate — none of them affect the ratio P_A / P_B .

Verification. Starting from the definition:

$$\frac{P_A}{P_B} = \frac{\exp(z_A) / Z}{\exp(z_B) / Z} = \frac{\exp(z_A)}{\exp(z_B)} = \exp(z_A - z_B)$$

The partition function Z cancels. The ratio depends only on the score difference $z_A - z_B$.

Physical meaning. The relative binding preference between two antigens is a property of the antibody-antigen pair, independent of what other antigens are in the pool. Adding a third antigen to the solution does not change the free energy difference between binding Antigen A and binding Antigen B. This follows from the fact that the binding free energy between two molecules is a property of that pair alone.

Preview of Lecture 6. The IIA property is what eliminates every candidate function except the exponential. The sigmoid, for example, fails IIA — its ratio depends on the absolute magnitudes of the scores, not just the difference. This is the decisive step in the uniqueness proof.

6. Softmax as “Soft” Argmax

The name “softmax” comes from its relationship to the argmax function.

Argmax (hard maximum). The argmax function assigns probability 1 to the highest score and 0 to all others:

$$\text{argmax}(z)_j = \begin{cases} 1 & \text{if } j = \arg \max_k z_k \\ 0 & \text{otherwise} \end{cases}$$

Argmax is a “winner-take-all” function. It violates Property 1 (positivity) because non-winners receive exactly zero probability.

Softmax as a smooth approximation. Softmax assigns the highest probability to the highest score but gives non-zero probability to all candidates. As the scores become more extreme (or, equivalently, as the temperature decreases — see Derivation Guide 5), softmax approaches argmax. In the opposite limit (all scores equal, or temperature very high), softmax produces a uniform distribution.

Worked comparison with $z = (2.0, 1.0, 0.1)$:

	$z_1 = 2.0$	$z_2 = 1.0$	$z_3 = 0.1$
Argmax	1.000	0.000	0.000
Softmax	0.659	0.242	0.099

Softmax preserves the ranking but distributes probability across all candidates. This is physically correct for molecular binding at non-zero temperature: the antibody preferentially binds its best target but has non-zero probability of binding weaker targets.

7. Worked Example 2: Negative and Mixed Scores

Problem. Compute softmax for $z = (3.0, -1.0, 0, -2.5)$.

This example demonstrates that softmax handles negative scores without difficulty (Property 4).

Step 1. Exponentiate each score:

$$\begin{aligned} \exp(3.0) &= 20.086 \\ \exp(-1.0) &= 0.368 \\ \exp(0) &= 1.000 \\ \exp(-2.5) &= 0.082 \end{aligned}$$

Step 2. Compute the partition function:

$$Z = 20.086 + 0.368 + 1.000 + 0.082 = 21.536$$

Step 3. Compute probabilities:

$$P_1 = \frac{20.086}{21.536} = 0.933$$

$$P_2 = \frac{0.368}{21.536} = 0.017$$

$$P_3 = \frac{1.000}{21.536} = 0.046$$

$$P_4 = \frac{0.082}{21.536} = 0.004$$

Step 4. Verify: $0.933 + 0.017 + 0.046 + 0.004 = 1.000$ ✓

Interpretation. Candidate 1 (score = 3.0) dominates with 93.3% of the probability. The negative-scoring candidates receive very little probability but are not excluded entirely. The rank order is preserved: $3.0 > 0 > -1.0 > -2.5$ maps to $0.933 > 0.046 > 0.017 > 0.004$.

8. The Numerical Stability Trick

In practice, computing $\exp(z_j)$ for large z_j causes numerical overflow (the number exceeds the range of floating-point representation). For example, $\exp(1000) \approx 10^{434}$, which overflows a 64-bit float.

The shift invariance property. Softmax is invariant to adding a constant c to all scores:

$$\text{softmax}(z+c)_j = \frac{\exp(z_j+c)}{\sum_k \exp(z_k+c)} = \frac{\exp(c)\exp(z_j)}{\exp(c)\sum_k \exp(z_k)} = \frac{\exp(z_j)}{\sum_k \exp(z_k)} = \text{softmax}(z)_j$$

The factor $\exp(c)$ cancels.

Practical algorithm. Before computing softmax, subtract the maximum score from all entries:

$$c = -\max_j z_j, \tilde{z}_j = z_j - \max_j z_j$$

Now $\max_j \tilde{z}_j = 0$, so $\exp(\tilde{z}_j) \leq 1$ for all j , eliminating overflow. The output is unchanged because of shift invariance.

Worked example. For $z = (1000, 999, 998)$:

Without the trick: $\exp(1000)$ overflows.

With the trick: $\tilde{z} = (0, -1, -2)$. Then $\exp(0) = 1$, $\exp(-1) = 0.368$, $\exp(-2) = 0.135$. $Z = 1.503$.

Softmax = $(0.665, 0.245, 0.090)$.

Physical interpretation of shift invariance. In the Boltzmann distribution, adding a constant to all energies does not change the probability distribution — only energy *differences* matter. This is equivalent to the freedom to choose a reference energy. The shift invariance of softmax is the same principle: only score differences determine the probability distribution.

9. Softmax and Simple Normalization

A natural question: why not simply normalize the scores by dividing by their sum?

Simple normalization:

$$P_j^{\text{simple}} = \frac{z_j}{\sum_k z_k}$$

Why this fails. Simple normalization violates Property 1 (positivity) when any score is negative: $z_j < 0$ implies $P_j < 0$, which is not a valid probability. It also violates Property 4 if scores can be negative. In molecular binding, unfavorable interactions have positive ΔG (negative scores after sign flip), so negative scores are physically required.

Worked comparison with $z = (3, -1, 0, -2.5)$:

	Simple normalization	Softmax
$z_1 = 3.0$	$3.0 / (-0.5) = -6.0$	0.933
$z_2 = -1.0$	$-1.0 / (-0.5) = 2.0$	0.017
$z_3 = 0$	$0 / (-0.5) = 0$	0.046
$z_4 = -2.5$	$-2.5 / (-0.5) = 5.0$	0.004

Simple normalization produces negative and rank-reversed “probabilities” (the most negative score gets the highest value). It is not a valid probability distribution. Softmax produces a valid distribution that preserves rank order.

The exponential in softmax is not an arbitrary choice — it is the function that simultaneously guarantees positivity, handles unrestricted inputs, preserves rank order, and satisfies IIA. In Lecture 6, we prove that it is the *only* such function.

10. Softmax with Equal Scores

Problem. Compute softmax for $z = (c, c, c, c)$ — four equal scores.

Step 1. Exponentiate: $\exp(c)$ for all four entries.

Step 2. Partition function: $Z = 4 \exp(c)$.

Step 3. Probabilities:

$$P_j = \frac{\exp(c)}{4 \exp(c)} = \frac{1}{4} = 0.25 \text{ for all } j$$

Result. When all scores are equal, softmax produces a **uniform distribution**. This holds regardless of the value of c — by shift invariance, only differences matter, and all differences are zero.

Physical meaning. If an antibody has equal binding energy with all antigens in the pool, it selects each with equal probability. There is no preference.

General formula. For M candidates with equal scores:

$$P_j = \frac{1}{M} \text{ for all } j$$

This is the maximum entropy distribution — the distribution that commits least to any particular outcome. In Lecture 4, we will see that this connects to Jaynes’ maximum entropy principle (1957): in the absence of any information favoring one candidate over another, the rational assignment is uniform.

11. How Softmax Amplifies Differences

The exponential function amplifies score differences non-linearly. A small additive difference in scores becomes a multiplicative difference in probabilities.

Formal statement. For two scores z_A and z_B with difference $\delta = z_A - z_B$:

$$\frac{P_A}{P_B} = \exp(\delta)$$

The probability ratio grows exponentially with the score difference.

Numerical examples:

Score difference δ	Probability ratio $\exp(\delta)$	Interpretation
0	1.0	Equal preference
1	2.72	A is 2.7× more likely
2	7.39	A is 7.4× more likely
3	20.09	A is 20× more likely
5	148.4	A is 148× more likely
10	22,026	A is 22,000× more likely

A score advantage of just 5 units produces a 148-fold probability advantage. This exponential amplification is a direct consequence of the Boltzmann distribution: in statistical mechanics, a free

energy difference of $5 k_B T$ between two states produces a $\exp(5) \approx 148$ -fold population difference. The same mathematics governs molecular selection stringency.

12. Summary

Concept	Definition	Significance
Softmax	$P_j = \exp(z_j) / \sum_k \exp(z_k)$	Converts scores to probabilities
Positivity	$P_j > 0$ for all j	Every candidate has non-zero probability
Normalization	$\sum P_j = 1$	Output is a valid probability distribution
Rank preservation	$z_A > z_B \Rightarrow P_A > P_B$	Ordering is never reversed
Unrestricted domain	$z_j \in \mathbb{R}$	Handles negative scores (unfavorable binding)
IIA	$P_A / P_B = \exp(z_A - z_B)$	Ratio depends only on score difference
Shift invariance	$\text{softmax}(z + c) = \text{softmax}(z)$	Only differences matter; free reference energy
Partition function	$Z = \sum_k \exp(z_k)$	Boltzmann's Zustandssumme
Equal scores	$P_j = 1 / M$	Uniform distribution = maximum entropy
Amplification	$P_A / P_B = \exp(\delta)$	Exponential in score difference

13. Checkpoint Questions

Q1. Compute softmax for $z = (1, 2, 3)$.

Answer: $\exp(1) = 2.718$, $\exp(2) = 7.389$, $\exp(3) = 20.086$. $Z = 30.193$. Softmax $= (0.090, 0.245, 0.665)$. The highest score (3) receives 66.5% of the probability.

Q2. An SFM produces compatibility scores of $s = (4.2, 4.0, 3.8, 2.1, 0.5)$ for five candidate targets. Without computing the full softmax, which candidate will receive the most probability, and approximately how much more likely is it than the second-best candidate?

Answer: Candidate 1 ($s = 4.2$) receives the most probability. The ratio $P_1 / P_2 = \exp(4.2 - 4.0) = \exp(0.2) \approx 1.22$. So Candidate 1 is about 22% more likely than Candidate 2. The small score difference (0.2) produces a modest probability advantage. Compare this to Candidate 1 vs. Candidate 5: $\exp(4.2 - 0.5) = \exp(3.7) \approx 40.4$. Candidate 1 is about 40× more likely than Candidate 5.

Q3. A student proposes using the function $P_j = z_j^2 / \sum_k z_k^2$ as an alternative to softmax. Which of the five properties does this violate?

Answer: This function satisfies positivity ($z_j^2 \geq 0$, and > 0 for $z_j \neq 0$), normalization (by construction), and unrestricted domain (z_j^2 is defined for all z_j). However, it violates **rank preservation**: for $z_A = -3$ and $z_B = 1$, we have $z_A < z_B$ but $z_A^2 = 9 > 1 = z_B^2$, so $P_A > P_B$. The function reverses the ranking for negative scores. It also violates **IIA**: $P_A / P_B = z_A^2 / z_B^2 = (z_A / z_B)^2$, which depends on the ratio of scores, not the difference.

Q4. Softmax is shift-invariant: adding a constant to all scores doesn't change the output. Is it also scale-invariant? That is, does $\text{softmax}(\alpha z) = \text{softmax}(z)$ for $\alpha > 0$?

Answer: No. Multiplying all scores by α changes the output.

$\text{softmax}(\alpha z)_j = \exp(\alpha z_j) / \sum_k \exp(\alpha z_k)$. This is equivalent to softmax with temperature $\tau = 1 / \alpha$:

multiplying scores by 2 is the same as halving the temperature (sharpening the distribution).

Multiplying by 0.5 is the same as doubling the temperature (flattening the distribution). This is derived formally in Derivation Guide 5.

14. Connection to Future Lectures

- **Derivation Guide 5 (next):** Introduces the temperature parameter τ , giving $\text{softmax}(z/\tau)$. Temperature controls the sharpness of selection and has a direct physical interpretation as $k_B T$.
- **Lecture 4:** The Boltzmann distribution $P(j) = \exp(-E_j/k_B T)/Z$ is softmax applied to negated energies divided by thermal energy. The partition function Z is derived from statistical mechanics via maximum entropy.
- **Lecture 6:** Luce's Choice Axiom (1959) proves that softmax is the *unique* function satisfying all five properties. Every other candidate function — sigmoid, power law, ReLU, polynomial — fails at least one property. The elimination proof is the mathematical centerpiece of the course.
- **Lecture 7:** InfoNCE, the training loss for SFMs, is derived as the negative log-likelihood of the softmax selection probability. The training objective IS the softmax function applied to the training scenario.
- **Lecture 9:** The MRC scaling law uses the softmax-derived selection probability to predict exponential convergence of retrieval accuracy with training data diversity.